NEET Chemistry Golden Set v1.0

90 curated mixed-style questions for RAG problem-solving evaluation

Instructions: This is a question-only version. Each question has four options.
Reaction schemes, chemical structures, equations, graphs and matching tables are preserved from the source qu
Answers, solutions, chapter headings, difficulty labels, and original source numbering are intentionally not inclu

Q1.

Among the following, choose the ones with equal number of atoms.

A. 212 g of Na_2CO_3 (s) [molar mass = 106 g]

B. 248 g of Na_2O (s) [molar mass = 62 g]

C. 240 g of NaOH (s) [molar mass = 40 g]

D. 12 g of H_2(g) [molar mass = 2 g]

E. 220 g of CO_2(g) [molar mass = 44 g]

Choose the **correct** answer from the options given below:

(1) A, B, and C only

(2) A, B, and D only

(3) B, C, and D only

(4) B, D, and E only

Q2.

In Lassaigne's extract of an organic compound, both nitrogen and sulphur are present, which gives blood red colour with Fe^{3+} due to the formation of

a. $[Fe(SCN)]^{2+}$

b. $Fe_4[Fe(CN)_6]_3.xH_2O$

c. NaSCN

d. $[Fe(CN)_5NOS]^{4-}$

Q3.

Which of the following statements are true?

A. Unlike Ga that has a very high melting point, Cs has a very low melting point.

B. On Pauling scale, the electronegativity values of N and Cl are not the same.

C. Ar, K^+, Cl^-, Ca^{2+} and S^{2-} are all isoelectronic species.

D. The correct order of the first ionization enthalpies of Na, Mg, Al, and Si

is Si > Aℓ > Mg > Na.

E. The atomic radius of Cs is greater than that of Li and Rb.

Choose the **correct** answer from the options given below:

(1) A, B and E only

(2) C and E only

(3) C and D only

(4) A, C and E only

Q4.

A compound X contains 32% of A, 20% of B and remaining percentage of C. Then, the empirical formula of X is:
(Given atomic masses of A = 64; B = 40; C = 32u)

(1) AB_2C_2 (2) ABC_4

(3) A_2BC_2 (4) ABC_3

Q5.

Arrange the following elements in increasing order of electronegativity:
N, O, F, C, Si
Choose the correct answer from the options given below:

(1) O < F < N < C < Si

(2) F < O < N < C < Si

(3) Si < C < N < O < F

(4) Si < C < O < N < F

Q6.

Match the **List-I** with **List-II.**

	List-I (Mixture)		**List-II (Method of Separation)**
A.	$CHCl_3$ + $C_6H_5NH_2$	I.	Distillation under reduced pressure
B.	Crude oil in petroleum industry	II.	Steam distillation
C.	Glycerol from spent-lye	III.	Fractional distillation
D.	Aniline-water	IV.	Simple distillation

Choose the **correct** answer from the options given below.

(1) A-IV, B-III, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-IV, C-II, D-I

Q7.

The highest number of helium atoms is in

(1) 4 g of helium

(2) 2.271098 L of helium at STP

(3) 4 mol of helium

(4) 4 u of helium

Q8.

Which one of the following reactions does **NOT** belong to "Lassaigne's test"?

(1) $Na + C + N \xrightarrow[\Delta]{} NaCN$

(2) $2Na + S \xrightarrow[\Delta]{} Na_2S$

(3) $Na + X \xrightarrow[\Delta]{} + NaX$

(4) $2CuO + C \xrightarrow[\Delta]{} 2Cu + CO_2$

Q9.

Arrange the following elements in increasing order of first ionization enthalpy:

Li, Be, B, C, N

Choose the correct answer from options given below:

(1) Li < Be < C < B < N

(2) Li < Be < N < B < C

(3) Li < Be < B < C < N

(4) Li < B < Be < C < N

Q10.

The energy of an electron in the ground state (n =1) for He^+ ion is –x J, then that for an electron in n = 2 state for Be^{3+} ion in J is:

(1) $-4x$

(2) $-\frac{4}{9}x$

(3) $-x$

(4) $-\frac{x}{9}$

Q11.

Which among the following electronic configurations belong to main group elements?

A. $[Ne]3s^1$ B. $[Ar]3d^3\ 4s^2$

C. $[Kr]4d^{10}5s^25p^5$ D. $[Ar]3d^{10}4s^1$

E. $[Rn]5f^06d^27s^2$

Choose the **correct** answer from the options given below:

(1) B and E only

(2) A and C only

(3) D and E only

(4) A, C and D only

Q12.

A compound with a molecular formula of C_6H_{14} has two tertiary carbons. Its IUPAC name is:

(1) 2,3-dimethylbutane

(2) 2,2-dimethylbutane

(3) n-hexane

(4) 2-methylpentane

Q13.

Energy and radius of first Bohr orbit of He^+ and Li^{2+} are [Given $R_H = 2.18\times10^{-18}$ J, $a_0 = 52.9$ pm]

(1) $E_n\,(Li^{2+}) = -19.62 \times 10^{-18}$ J:
$r_n\,(Li^{2+}) = 17.6$ pm
$E_n\,(He^+) = -8.72\times 10^{-18}$ J;
$r_n\,(He^+) = 26.4$ pm

(2) $E_n\,(Li^{2+}) = -8.72 \times 10^{-18}$ J;
$r_n\,(Li^{2+}) = 26.4$ pm
$E_n\,(He^+) = -19.62 \times 10^{-18}$ J:
$r_n\,(He^+) = 17.6$ pm

(3) $E_n\,(Li^{2+}) = -19.62 \times 10^{-16}$ J;
$r_n\,(Li^{2+}) = 17.6$ pm
$E_n\,(He^+) = -8.72 \times 10^{-16}$ J:
$r_n\,(He^+) = 26.4$ pm

(4) $E_n\,(Li^{2+}) = -8.72 \times 10^{-16}$ J;
$r_n\,(Li^{2+}) = 17.6$ pm
$E_n\,(He^{2+}) = -19.62 \times 10^{-16}$ J:
$r_n\,(He^+) = 17.6$ pm

Q14.

Given below are two statements:

Statement I : The boiling point of three isomeric pentanes follows the order.

n-pentane > isopentane > neopentane

Statement II : When branching increases, the molecule attains a shape of sphere. This results in smaller surface area for contact, due to which the intermolecular forces between the spherical molecules are weak, thereby lowering the boiling point.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

Q15.

Match List-I with List-II.

	List-I (Compound)		**List-II (Shape/geometry)**
A.	NH_3	I.	Trigonal Pyramidal
B.	BrF_5	II.	Square Planar
C.	XeF_4	III.	Octahedral
D.	SF_6	IV.	Square Pyramidal

Choose the correct answer from the options given below:

(1) A-III, B-IV, C-I, D-II

(2) A-II, B-III, C-IV, D-I

(3) A-I, B-IV, C-II, D-III

(4) A-II, B-IV, C-III, D-I

Q16.

Select the correct statements from the following:

A. Atoms of all elements are composed of two fundamental particles.

B. The mass of the electron is 9.10939×10^{-31} kg.

C. All the isotopes of a given element show same chemical properties.

D. Protons and electrons are collectively known as nucleons.

E. Dalton's atomic theory, regarded the atom as an ultimate particle of matter.

Choose the correct answer from the options given below:

a. B, C and E only

b. A, B and C only

c. C, D and E only

d. A and E only

Q17.

Match the **List-I** with **List-II.**

	List-I		**List-II**
A.	XeO_3	I.	sp^3d; linear
B.	XeF_2	II.	sp^3; pyramidal
C.	$XeOF_4$	III.	sp^3d^3: distorted octahedral
D.	XeF_6	IV.	sp^3d^2; square pyramidal

Choose the **correct** answer from the options given below.

(1) A-II, B-I, C-IV, D-III

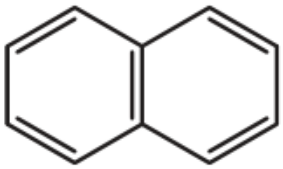
(2) A-II, B-I, C-III, D-IV

(3) A-IV, B-II, C-III, D-I

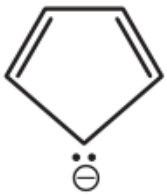
(4) A-IV, B-II, C-I, D-III

Q18.

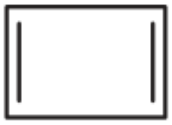
Consider the following compounds/species :



(I)



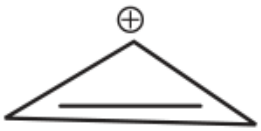
(II)



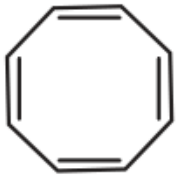
(III)



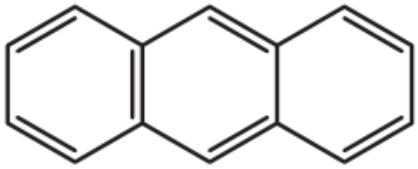
(IV)



(V)



(VI)



(VII)

The number of compounds/species which obey Huckel's rule is _____.

a. 5
b. 4
c. 6
d. 2

Environmental Chemistry

Q19.

1 gram of sodium hydroxide was treated with 25 mL of 0.75 M HCl solution, the mass of sodium hydroxide left unreacted is equal to

(1) Zero mg
(2) 200 mg
(3) 750 mg
(4) 250 mg

Q20.

Which one of the following compounds can exist as cis-trans isomers?

(1) Pent-1-ene

(2) 2-Methylhex-2-ene

(3) 1,1-Dimethylcyclopropane

(4) 1.2-Dimethylcyclohexane

Q21.

Identify the correct orders against the property mentioned

A. $H_2O > NH_3 > CHCl_3$ - dipole moment

B. $XeF_4 > XeO_3 > XeF_2$ – number of lone pairs on central atom

C. O–H > C–H > N–O – bond length

D. $N_2 > O_2 > H_2$ – bond enthalpy

Q22.

Consider the following compounds:

$\underline{K}O_2$, $H_2\underline{O}_2$ and $H_2\underline{S}O_4$

The oxidation states of the underlined elements in them are, respectively,

(1) +1, –1, and +6

(2) +2, –2, and +6

(3) +1, –2, and +4

(4) +4, –4, and +6

Q23.

Given below are two statements:

Statement I: A hypothetical diatomic molecule with bond order zero is quite stable.

Statement II: As bond order increases, the bond length increases.

In the light of the above statements, chose the **most appropriate** answer from the options given below:

(1) Both statement I and Statement II are true

(2) Both statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is true but Statement II is true

Q24.

How many products (including stereoisomers) are expected from monochlorination of the following compound?



(1) 2 (2) 3

(3) 5 (4) 6

Q25.

In which of the following processes entropy increases?

A. A liquid evaporates to vapour.

B. Temperature of a crystalline solid lowered from 130 K to 0 K.

C. $2NaHCO_{3(s)} \rightarrow Na_2CO_{3(s)} + CO_{2(g)} + H_2O_{(g)}$

D. $Cl_{2(g)} \rightarrow 2Cl_{(g)}$

Choose the correct answer from the options given below:

(1) A, C and D

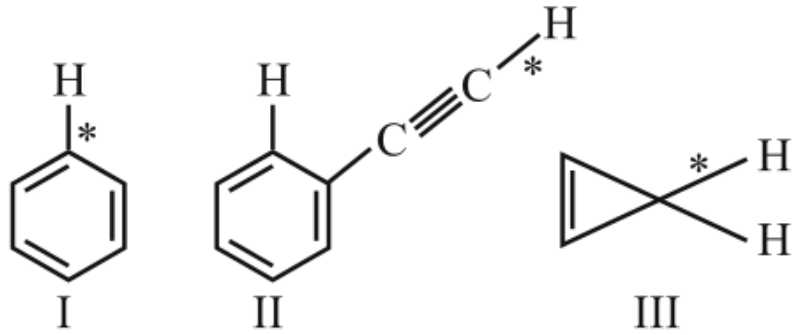
(2) C and D

(3) A and C

(4) A, B and D

Q26.

Among the given compounds I-III, the correct order of bond dissociation energy of C–H bond marked with * is:



(1) II > I > III

(2) I > II > III

(3) III > II > I

(4) II > III > I

Q27.

Amongst the following, the total number of species NOT having eight electrons around central atom in its outer most shell, is: NH_3, $AlCl_3$, $BeCl_2$, CCl_4, PCl_5

a. 1

b. 3

c. 2

d. 4

Q28.

The work done during reversible isothermal expansion of one mole of hydrogen gas at 25°C from pressure of 20 atmosphere to 10 atmosphere is:
(Given R = 2.0 cal $K^{-1}mol^{-1}$)

(1) 413.14 calories

(2) 100 calories

(3) 0 calorie

(4) – 413.14 calories

Q29.

Identify the **correct** answer.

(1) Dipole moment of NF_3 is greater than that of NH_3.

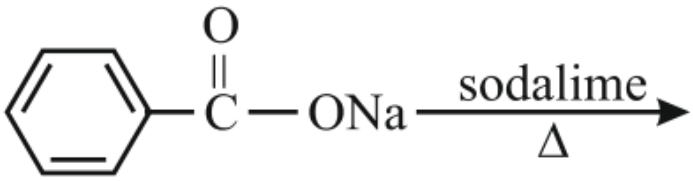
(2) Three canonical forms can be drawn for CO_3^{2-} ion.

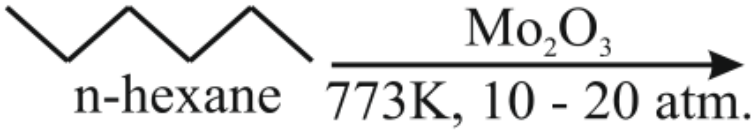
(3) Three resonance structures can be drawn for ozone.

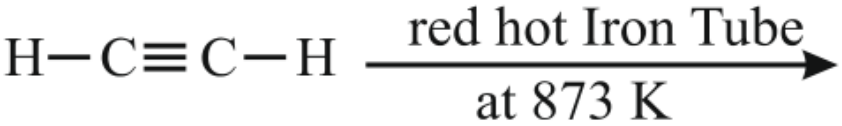
(4) BF_3 has non-zero dipole moment.

Q30.

Which one of the following reactions does **NOT** give benzene as the product?

(1) 

(2) 

(3) 

(4) 

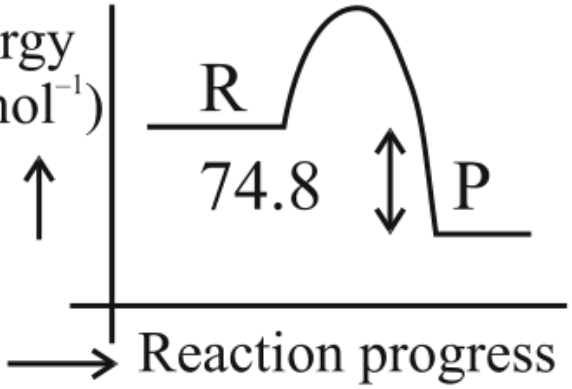
Q31.

$C(s) + 2H_2(g) \rightarrow CH_4(g);\ \Delta H = -74.8\ kJ\ mol^{-1}$

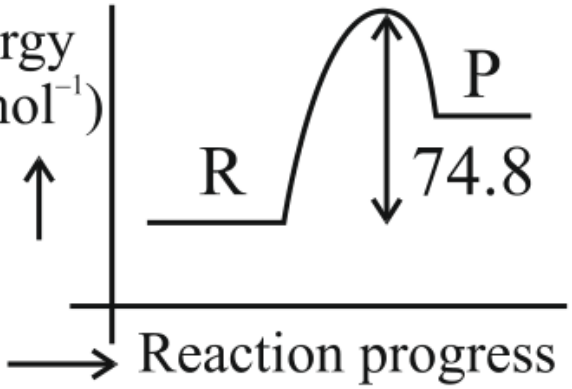
Which of the following diagrams gives an accurate representation of the above reaction?

[R → reactants; P → products]

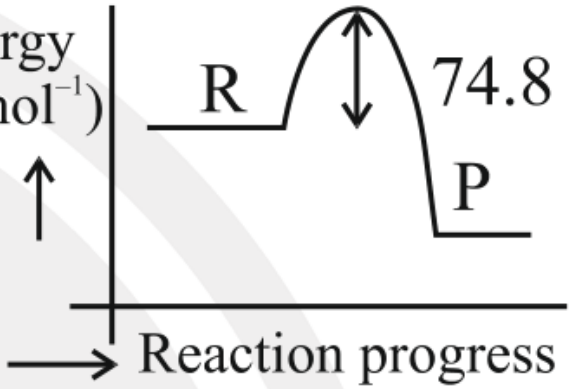
(1)



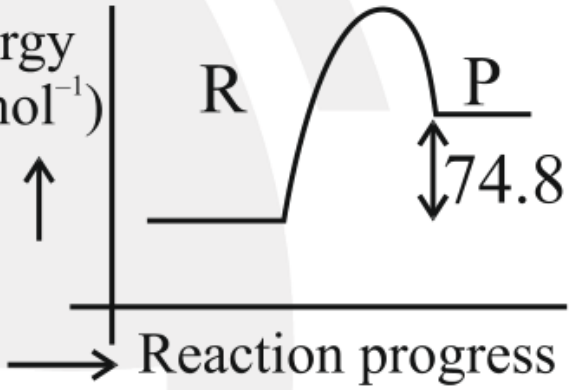
(2)



(3)

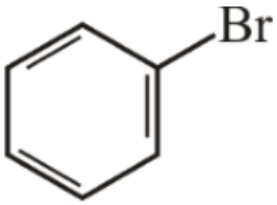


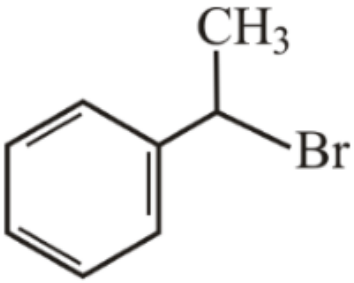
(4)

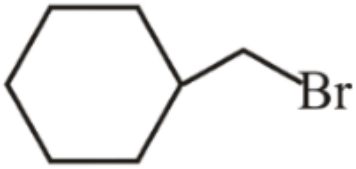


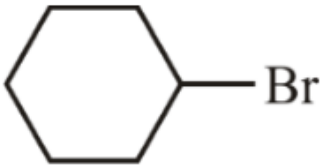
Q32.

The compound that will undergo S_N^1 reaction with the fastest rate is

(1) 

(2) 

(3) 

(4) 

Q33.

Given below are two statements:

Statement I: Like nitrogen that can form ammonia, arsenic can form arsine.

Statement II: Antimony cannot form antimony pentoxide.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

Q34.

The standard heat of formation, in kcal/mol of Ba^{2+} is:

[Given : standard heat of formation of SO_4^{2-} ion (aq)= –216 kcal/mol, standard heat of crystallization of $BaSO_4(s)$ = –4.5 kcal/mol, standard heat of formation of $BaSO_4(s)$ = –349 kcal/mol]

(1) –128.5

(2) –133.0

(3) +133.0

(4) +220.5

Q35.

Given below are two statements:

Statement I : The boiling point of hydrides of Group 16 elements follow the order

$H_2O > H_2Te > H_2Se > H_2S$.

Statement II : On the basis of molecular mass, H_2O is expected to have lower boiling point than the other members of the group but due to the presence of extensive H-bonding in H_2O, it has higher boiling point.

In the light of the above statements, choose the *correct* answer from the options given below:

(1) Statement I is true but Statement II is false.
(2) Statement I is false but Statement II is true.
(3) Both Statement I and Statement II are true.
(4) Both Statement I and Statement II are false.

Q36.

Given below are two statement: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**:

Assertion (A):  undergoes S_N2 reaction faster than  .

Reason (R): Iodine is a better leaving group because of its large size.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both **A** and **R** are true and **R** is the correct explanation of **A**.

(2) Both **A** and **R** are true but **R** is not the correct explanation of **A**.

(3) **A** is true but **R** is false.

(4) **A** is false but **R** is true.

Q37.

For the reaction $2A \rightleftharpoons B + C$, $K_c = 4 \times 10^{-3}$. At a given time, the composition of reaction mixture is: $[A] = [B] = [C] = 2 \times 10^{-3}$ M.

Then, which of the following is correct?

(1) Reaction has a tendency to go in backward direction.

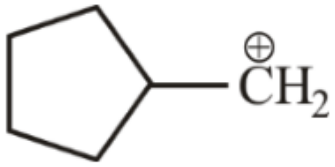
(2) Reaction has gone to completion in forward direction.

(3) Reaction is at equilibrium.

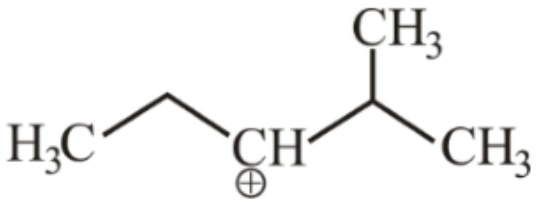
(4) Reaction has a tendency to go in forward direction.

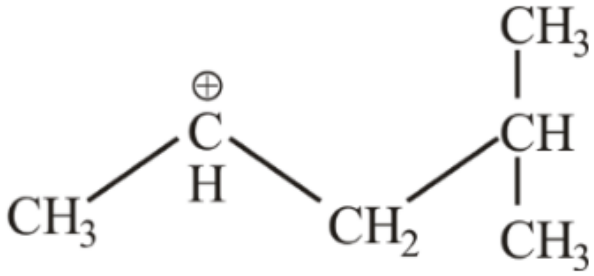
Q38.

The most stable carbocation among the following is:

(1) 

(2) 

(3) 

(4) 

Q39.

Among Group 16 elements, which one does **NOT** show –2 oxidation state?

(1) Te

(2) Po

(3) O

(4) Se

Q40.

Consider the following reaction in a sealed vessel at equilibrium with concentrations of
$N_2 = 3.0 \times 10^{-3}$ M, $O_2 = 4.2 \times 10^{-3}$ M and NO = 2.8×10^{-3} M.

$$2NO_{(g)} \rightleftharpoons N_{2(g)} + O_{2(g)}$$

If 0.1 mol L^{-1} of $NO_{(g)}$ is taken in a closed vessel, what will be degree of dissociation (α) of $NO_{(g)}$ at equilibrium?

(1) 0.8889
(2) 0.717
(3) 0.00889
(4) 0.0889

Q41.

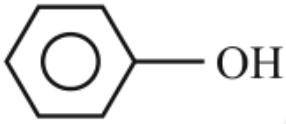
Taking stability as the factor, which one of the following represents correct relationship?

a. $TlI > TlI_3$
b. $TlCl_3 > TlCl$
c. $InI_3 > InI$
d. $AlCl > AlCl_3$

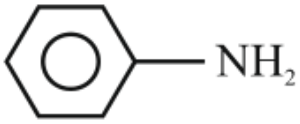
Q42.

Which one of the following compounds **does not** decolourize bromine water?

(1) 

(2) 

(3) 

(4) 

Q43.

Phosphoric acid ionizes in three steps with their ionization constant values K_{a_1}, K_{a_2} and K_{a_3}, respectively, while K is the overall ionization constant. Which of the following statements are true?

A. $\log K = \log K_{a_1} + \log K_{a_2} + \log K_{a_3}$

B. H_3PO_4 is stronger acid than $H_2PO_4^-$ and HPO_4^{2-}.

C. $K_{a_1} > K_{a_2} > K_{a_3}$

D. $K_{a_1} = \frac{K_{a_3} + K_{a_2}}{2}$

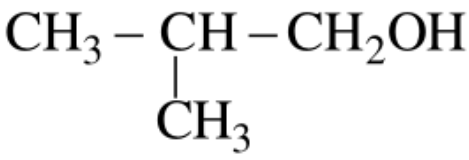
Choose the **correct** answer from the options given below:

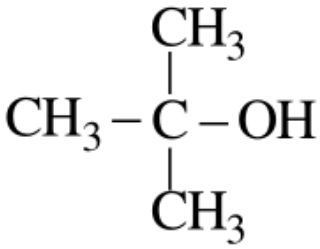
(1) A and B only (2) A and C only

(3) B, C and D only (4) A, B and C only

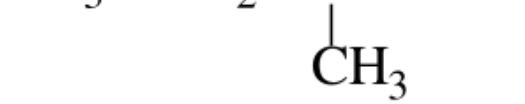
Q44.

Which one of the following alcohols reacts instantaneously with Lucas reagent?

(1) 

(2) 

(3) 

(4) 

Q45.

Match List-I with List-II:

	List-I (Oxoacids of Sulphur)		**List-II (Bonds of Sulphur)**
A.	Peroxodisulphuric acid	I.	Two S-OH, Four S=O, One S-O-S
B.	Sulphuric acid	II.	Two S-OH, One S=O
C.	Pyrosulphuric acid	III.	Two S-OH, Four S=O,One S-O-O-S
D.	Sulphurous acid	IV.	Two S-OH, Two S=O

Choose the correct answer from the options given below :

a. A-III, B-IV, C-II, D-I
b. A-I, B-III, C-II, D-IV
c. A-III, B-IV, C-I, D-II
d. A-I, B-III, C-IV, D-II

Q46.

For the reaction $A(g) \rightleftharpoons 2B(g)$, the backward reaction rate constant is higher than the forward reaction rate constant by a factor of 2500, at 1000 K.
[Given: R= 0.0831 L atm mol^{-1} K^{-1}]
K_P for the reaction at 1000 K is

(1) 83.1
(2) 2.077×10^5
(3) 0.033
(4) 0.021

Q47.

The E° value for the Mn^{3+}/Mn^{2+} couple is more positive than that of Cr^{3+}/Cr^{2+} or Fe^{3+}/Fe^{2+} due to change of

(1) d^4 to d^5 configuration

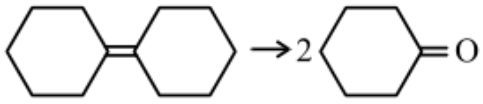
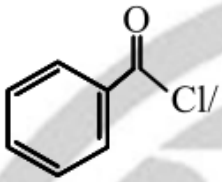
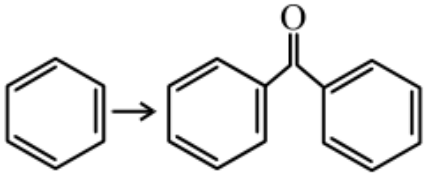
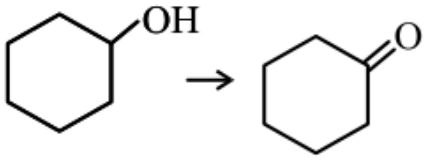
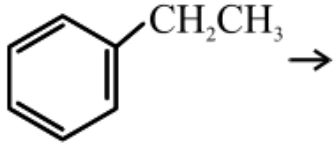
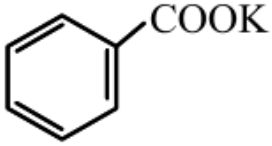
(2) d^3 to d^5 configuration

(3) d^5 to d^4 configuration

(4) d^5 to d^2 configuration

Q48.

Match List I with List II.

Lits I (Reaction)	**List II (Reagents/ Condition)**
A. 	I.  Anhyd.$AlCl_3$
B. 	II. CrO_3
C. 	III. $KMnO_4$/ KOH, Δ
D.  	IV. (i) O_3 (ii) Zn-H_2O

Choose the correct answer from the options given below:

(1) A-IV, B-I, C-II, D-III

(2) A-I, B-IV, C-II, D-III

(3) A-IV, B-I, C-III, D-II

(4) A-III, B-I, C-II, D-IV

Q49.

Higher yield of NO in

$N_2(g) + O_2 \rightleftharpoons 2NO(g)$ can be obtained at

[ΔH of the reaction = + 180.7 kJ mol^{-1}]

A. higher temperature

B. lower temperature

C. higher concentration of N_2

D. higher concentration of O_2

Choose the **correct** answer from the options given below:

(1) A, D only

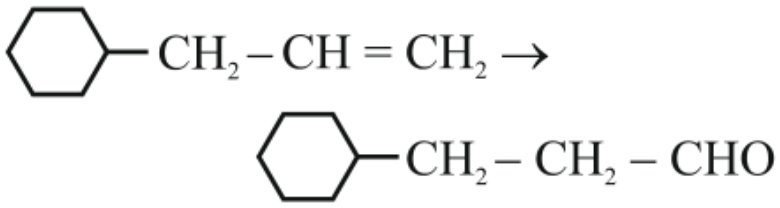
(2) B, C only

(3) B, C, D only

(4) A, C, D only

Q50.

Identify the correct reagents that would bring about the following transformation.



(1) (i) BH_3

(ii) $H_2O_2/\overset{\ominus}{O}H$

(iii) alk. $KMnO_4$

(iv) $H_3O^{\oplus}$

(2) (i) H_2O/H^+

(ii) PCC

Q51.

'Spin only' magnetic moment is same for which of the following ions?

A. Ti^{3+} B. Cr^{2+}

C. Mn^{2+} D. Fe^{2+}

E. Sc^{3+}

Choose the most appropriate answer from the options given below:

(1) B and C only (2) A and D only

(3) B and D only (4) A and E only

Q52.

The plot of osmotic pressure (Π) vs concentration (mol L^{-1}) for a solution gives a straight line with slope 25.73 L bar mol^{-1}. The temperature at which the osmotic pressure measurement is done is:
(Use R = 0.083 L bar mol^{-1} K^{-1})

(1) 25.73°C (2) 12.05°C
(3) 37°C (4) 310°C

Q53.

Given below are two statements;

Statement I: Ferromagnetism is considered as an extreme form of paramagnetism.

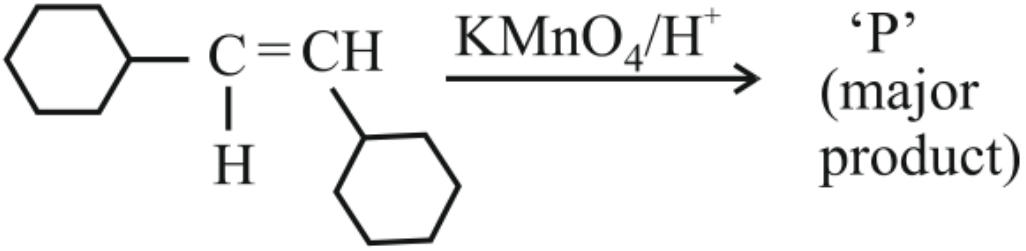
Statement II: The number of unpaired electrons in a Cr^{2+} ion (Z=24) is the same as that of a Nd^{3+} ion (Z= 60).

In the light of the above statements, choose the **correct** answer from the options given below:

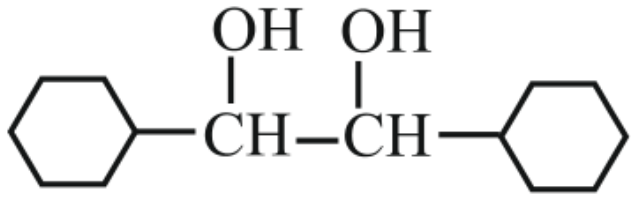
(1) Both Statement I and Statement II are true
(2) Both Statement I and Statement II are false
(3) Statement I is true but Statement II is false
(4) Statement I is false but Statement II is true

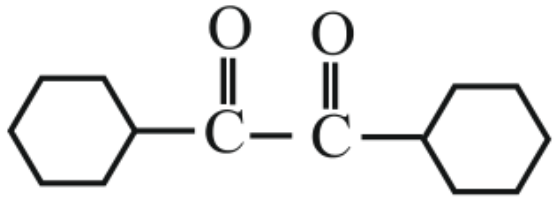
Q54.

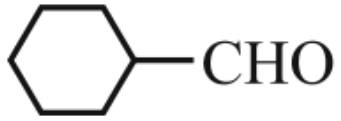
For the given reaction:

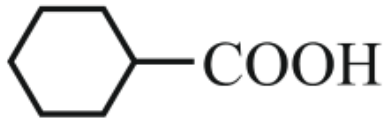


'P' is

(1) 

(2) 

(3) 

(4) 

Q55.

Which of the following aqueous solution will exhibit highest boiling point?

(1) 0.01 M Urea

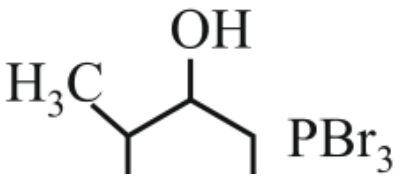
(2) 0.01 M KNO_3

(3) 0.01 M Na_2SO_4

(4) 0.015 M $C_6H_{12}O_6$

Q56.

Major products A and B formed in the following reaction sequence, are



Q57.

The pair of lanthanoid ions which are diamagnetic is

(1) Gd^{3+} and Eu^{3+}

(2) Pm^{3+} and Sm^{3+}

(3) Ce^{4+} and Yb^{2+}

(4) Ce^{3+} and Eu^{2+}

Q58.

5 moles of liquid X and 10 moles of liquid Y make a solution having a vapour pressure of 70 torr. The vapour pressures of pure X and Y are 63 torr and 78 torr respectively. Which of the following is true regarding the described solution?

(1) The solution shows positive deviation.

(2) The solution shows negative deviation.

(3) The solution is ideal.

(4) The solution has volume greater than the sum of individual volumes.

Q59.

Which of the following statements are INCORRECT?

A. All the transition metals except scandium form MC oxides which are ionic.

B. The highest oxidation number corresponding to the group number in transition metal oxides is attained in Sc_2O_3 tc Mn_2O_7.

C. Basic character increases from V_2O_3 to V_2O_4 to V_2O_5.

D. V_2O_4 dissolves in acids to give VO_4^{3-} salts.

E. CrO is basic but Cr_2O_3 is amphoteric.

Choose the correct answer from the options given below:

a. B and C only

b. A and E only

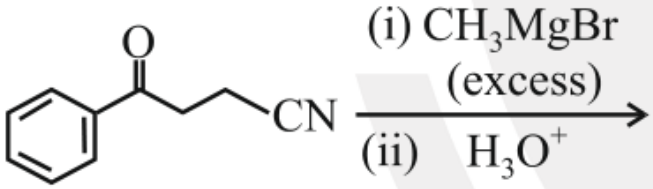
c. B and D only

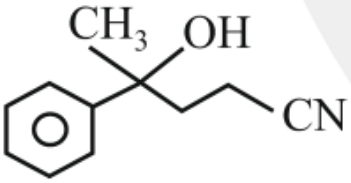
d. C and D only

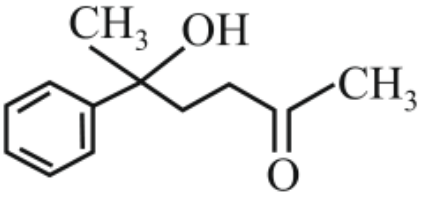
Coordination Compounds

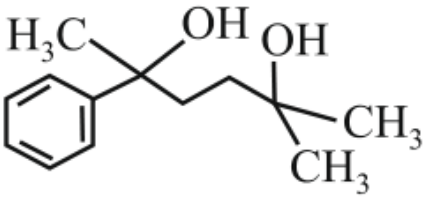
Q60.

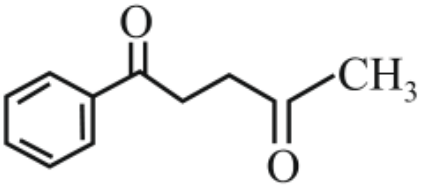
The major product of the following reaction is:



(1) 

(2) 

(3) 

(4) 

Q61.

Mass in grams of copper deposited by passing 9.6487 A current through a voltmeter containing copper sulphate solution for 100 seconds is:

(Given: Molar mass of Cu : 63 g mol^{-1}, 1F = 96487 C)

(1) 31.5 g

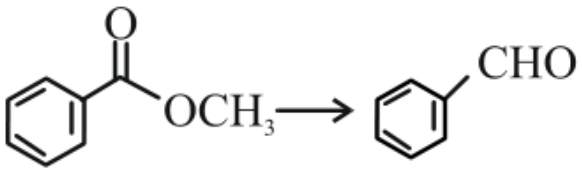
(2) 0.0315 g

(3) 3.15 g

(4) 0.315 g

Q62.

Identify the suitable reagent for the following conversion.



(1) (i) $LiAlH_4$, (ii) H^+/H_2O

(2) (i) $AlH(iBu)_2$ (ii) H_2O

(3) (i) $NaBH_4$, (ii) H^+/H_2O

(4) (i) $H_2/Pd\text{-}BaSO_4$

Q63.

Given below are two statements:

Statement I: Both $\left[Co(NH_3)_6\right]^{3+}$ and $\left[CoF_6\right]^{3-}$ complexes are octahedral but differ in their magnetic behaviour.

Statement II: $\left[Co(NH_3)_6\right]^{3+}$ is diamagnetic whereas $\left[CoF_6\right]^{3-}$ is paramagnetic.

In the light of the above statements, choose the *correct* answer from the options given below:

(1) Statement I is true but Statement II is false.

(2) Statement I is false but Statement II is true.

(3) Both Statement I and Statement II are true.

(4) Both Statement I and Statement II are false.

Q64.

If the molar conductivity $\left(\Lambda_m\right)$ of a 0.050 mol L^{-1} solution of a monobasic weak acid is 90 S cm^2 mol^{-1}, its extent (degree) of dissociation will be

[Assume $\Lambda^{\circ}_{+} = 349.6$ S cm^2 mol^{-1} and $\Lambda^{\circ}_{-} = 50.4$ S cm^2 mol^{-1}]

(1) 0.115

(2) 0.125

(3) 0.225

(4) 0.215

Q65.

Match List-I with List-II

List-I (Complex)	**List-II (Type of isomerism)**
A. $[Co(NH_3)_5(NO_2)]Cl_2$	I. Solvate isomerism
B. $[Co(NH_3)_5(SO_4)]Br$	II. Linkage isomerism
C. $[Co(NH_3)_6][Cr(CN)_6]$	III. Ionization isomerism
D. $[Co(H_2O)_6]Cl_3$	IV. Coordination isomerism

Choose the correct answer from the options given below:

(1) A-I, B-IV, C-III, D-II

(2) A-II, B-IV, C-III, D-I

(3) A-II, B-III, C-IV, D-I

(4) A-I, B-III, C-IV, D-II

Q66.

The reagents with which glucose does **not** react to give the corresponding tests/products are

A. Tollen's reagent

B. Schiff's reagent

C. HCN

D. NH_2OH

E. $NaHSO_3$

Choose the correct options from the given below:

(1) B and E

(2) E and D

(3) B and C

(4) A and D

Q67.

The conductivity of centimolar solution of KCl at 25°C is 0.0210 ohm^{-1} cm^{-1} and the resistance of the cell containing the solution at 25 °C is 60 ohm. The value of cell constant is:

a. 3.34 cm^{-1}

b. 1.34 cm^{-1}

c. 3.28 cm^{-1}

d. 1.26 cm^{-1}

Q68.

Sugar 'X'

A. is found in honey.

B. is a keto sugar.

C. exists in α and β – anomeric forms.

D. is laevorotatory.

'X' is:

(1) D-Glucose

(2) D-Fructose

(3) Maltose

(4) Sucrose

Q69.

The correct order of the wavelength of light absorbed by the following complexes is,

A. $[Co(NH_3)_6]^{3+}$ B. $[Co(CN)_6]^{3-}$

C. $[Cu(H_2O)_4]^{2+}$ D. $[Ti(H_2O)_6]^{3+}$

Choose the **correct** answer from the options given below:

(1) B < D < A < C (2) B < A < D < C

(3) C < D < A < B (4) C < A < D < B

Q70.

The rate of a reaction quadruples when temperature changes from 27°C to 57°C. Calculate the energy of activation.

Given R = 8.314 J K^{-1} mol^{-1}, log 4 = 0.6021

(1) 3.80 kJ/mol

(2) 3804 kJ/mol

(3) 38.04 kJ/mol

(4) 380.4 kJ/mol

Q71.

Out of the following complex compounds, which of the compound will be having the minimum conductance is solution?

(1) $[Co(NH_3)_3Cl_3]$

(2) $[Co(NH_3)_4Cl_2]$

(3) $[Co(NH_3)_6]Cl_3$

(4) $[Co(NH_3)_5Cl]Cl$

Q72.

Total number of possible isomers (both structural as well as stereoisomers) of cyclic ethers of molecular formula C_4H_8O is:

(1) 6

(2) 8

(3) 10

(4) 11

Q73.

If the half-life ($t_{1/2}$) for a first order reaction is 1 minute, then the time required for 99.9% completion of the reaction is closet to:

(1) 2 minutes (2) 4 minutes

(3) 5 minutes (4) 10 minutes

Q74.

Given below are two statements:

Statement I: Aniline does not undergo Friedel-Crafts alkylation reaction.

Statement II: Aniline cannot be prepared through Gabriel synthesis.

In the light of the above statements, choose the *correct* answer from the options given below:

(1) Statement I is correct but Statement II is false.

(2) Statement I is incorrect but Statement II is true.

(3) Both Statement I and Statement II are true.

(4) Both Statement I and Statement II are false.

Q75.

Which of the following are paramagnetic?

A. $[NiCl_4]^{2-}$ B. $Ni(CO)_4$

C. $[Ni(CN)_4]^{2-}$ D. $[Ni(H_2O)_6]^{2+}$

E. $Ni\ (PPh_3)_4$

Choose the **correct** answer from the options given below:

(1) A and C only (2) B and E only

(3) A and D only (4) A, D and E only

Q76.

If the rate constant of a reaction is 0.03 s^{-1}, how much time does it take for 7.2 mol L^{-1} concentration of the reactant to get reduced to 0.9 mol L^{-1}?

(Given : log 2 = 0.301)

(1) 69.3 s

(2) 23.1 s

(3) 210 s

(4) 21.0 s

Q77.

Given below are two statements:

Statement I: $[Co(NH_3)_6]^{3+}$ is a homoleptic complex whereas $[Co(NH_3)_4Cl_2]^{+}$ is a heteroleptic complex.

Statement II: Complex $[Co(NH_3)_6]^{3+}$ has only one kind of ligands but $[Co(NH_3)_4Cl_2]^{+}$ has more than one kind of ligands.

In the light of the above statements, choose the *correct* answer from the options given below:

(1) Statement I is true but Statement II is false.

(2) Statement I is false but Statement II is true.

(3) Both Statement I and Statement II are true.

(4) Both Statement I and Statement II are false.

Q78.

The correct order of decreasing basic strength of the given amines is:

(1) N-methylaniline > benzenamine > ethanamine > N-ethylethanamine

(2) N-ethylethanamine > ethanamine > benzenamine > N-methylaniline

(3) N-ethylethanamine > ethanamine > N-methylaniline > benzenamine

(4) benzenamine > ethanamine > N-methylaniline > N-ethylethanamine

Q79.

Activation energy of any chemical reaction can be calculated if one knows the value of

(1) orientation of reactant molecules during collision.

(2) rate constant at two different temperatures.

(3) rate constant at standard temperature.

(4) probability of collision.

Q80.

Identify the major product C formed in the following reaction sequence:

$$CH_3-CH_2-CH_2-I \xrightarrow{NaCN} A \xrightarrow[\text{Partial hydrolysis}]{OH^-} B \xrightarrow[Br_2]{NaOH} \underset{\text{(major)}}{C}$$

(1) butanamide
(2) α-bromobutanoic acid
(3) propylamine
(4) butylamine

Q81.

During the preparation of Mohr's salt solution (Ferrous ammonium sulphate), which of the following acid is added to prevent hydrolysis of Fe^{2+} ion?

(1) dilute nitric acid
(2) dilute sulphuric acid
(3) dilute hydrochloric acid
(4) concentrated sulphuric acid

Q82.

What fraction of one edge centred octahedral void lies in one unit cell of fcc?

a. $\frac{1}{12}$

b. $\frac{1}{2}$

c. $\frac{1}{3}$

d. $\frac{1}{4}$

Surface Chemistry

Q83.

Match **List-I** with **List-II.**

	List-I (Ion)		**List-II (Group Number in Cation Analysis)**
A.	Co^{2+}	I.	Group-I
B.	Mg^{2+}	II.	Group-III
C.	Pb^{2+}	III.	Group-IV
D.	$A\ell^{3+}$	IV.	Group-VI

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-II, D-I

(2) A-III, B-IV, C-I, D-II

(3) A-III, B-II, C-IV, D-I

(4) A-III, B-II, C-I, D-IV

Q84.

The correct order of decreasing acidity of the following aliphatic acids is:

(1) $(CH_3)_3CCOOH > (CH_3)_2CHCOOH > CH_3COOH > HCOOH$

(2) $CH_3COOH > (CH_3)_2CHCOOH > (CH_3)_3CCOOH > HCOOH$

(3) $HCOOH > CH_3COOH > (CH_3)_2CHCOOH > (CH_3)_3CCOOH$

(4) $HCOOH > (CH_3)_3CCOOH > (CH_3)_2CHCOOH > CH_3COOH$

Q85.

The Henry's law constant (K_H) values of three gases (A, B, C) in water are 145, 2×10^{-5} and 35 kbar, respectively. The solubility of these gases in water follow the order:

(1) A > C > B

(2) A > B > C

(3) B > A > C

(4) B > C > A

SECTION-B

Q86.

Match the **List-I** with **List-II.**

List-I (Name of Vitamin)		**List-II (Deficiency disease)**	
A.	Vitamin B_{12}	I.	Cheilosis
B.	Vitamin D	II.	Convulsions
C.	Vitamin B_2	III.	Rickets
D.	Vitamin B_6	.IV.	Pernicious anaemia

Choose the **correct** answer from the options given below.

(1) A-I, B-III, C-II, D-IV

(2) A-IV, B-III, C-I, D-II

(3) A-II, B-III, C-I, D-IV

(4) A-IV, B-III, C-II, D-I

Q87.

Given below are certain cations. Using inorganic qualitative analysis, arrange them in increasing group number from 0 to VI.

A. Al^{3+} B. Cu^{2+}

C. Ba^{2+} D. Co^{2+}

E. Mg^{2+}

Choose the correct answer from the option given below:

(1) E, C, D, B, A

(2) E, A, B, C, D

(3) B, A, D, C, E

(4) B, C, A, D, E

Q88.

Match the **List-I** with **List-II.**

List-I (Example)		**List-II (Type of Solution)**	
A.	Humidity	I.	Solid in solid
B.	Alloys	II.	Liquid in gas
C.	Amalgams	III.	Solid in gas
D.	Smoke	IV.	Liquid in solid

Choose the **correct** answer from the options given below.

(1) A-II, B-IV, C-I, D-III

(2) A-II, B-I, C-IV, D-III

(3) A-III, B-I, C-IV, D-II

(4) A-III, B-II, C-I, D-IV

Q89.

Match **List-I** with **List-II**.

	List-I		**List-II**
A.	Haber process	I.	Fe catalyst
B.	Wacker oxidation	II.	$PdCl_2$
C.	Wilkinson catalyst	III.	$[(PPh_3)_3RhCl]$
D.	Ziegler catalyst	IV.	$TiCl_4$ with $Al(CH_3)_3$

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-IV, D-III

(2) A-II, B-III, C-I, D-IV

(3) A-I, B-II, C-III, D-IV

(4) A-I, B-IV, C-III, D-II

Q90.

Fehling's solution 'A' is

(1) alkaline solution of sodium potassium tartrate (Rochelle's salt)

(2) aqueous sodium citrate

(3) aqueous copper sulphate

(4) alkaline copper sulphate